

## **Clubhouse: *ItsAboutTime.club***

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                               | CH Title                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Listen                    |
|---------------------------------------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">Ken Wharton</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>   | Bell's Theorem and Retrocausality                                 | Combined with quantum entanglement experiments, Bell's Theorem has long proved that we can't have nice things when it comes to space, time, and causality. But it is now becoming widely appreciated that we can still demand locally-mediated influences if we are willing to allow for hidden retrocausality. This talk will describe the various motivations for developing retrocausal models of quantum phenomena, detail some promising models already in the literature, and summarize their prospects for restoring a spacetime-based reality to fundamental physics. | <a href="#">4/22/2023</a> |
| No Session Today                                                                      | No Session Today                                                  | Easter Break (April 8th/9th). And Quantum Technologies Meetup.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 8-Apr-23                  |
| <a href="#">Pat Helland</a>                                                           | Reducing Jitter in Cloud Databases                                | Gray failures in clouds cause delays in responses to requests. Can we build fast transactional DBs if ENOUGH servers answer quickly while others are slow? What does time mean for transactions?                                                                                                                                                                                                                                                                                                                                                                              | <a href="#">4/1/2023</a>  |
| <a href="#">David Wolpert</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | Computational Complexity Theory and Thermodynamic Resource Costs. | Resource costs in Computational complexity depend on the size of the input. We summarize recent results of its relationship to heat, and research issues at the intersection of stochastic thermodynamics and computer science theory.                                                                                                                                                                                                                                                                                                                                        | <a href="#">3/25/2023</a> |
| <a href="#">Edward A. Lee</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | Information, Entropy, and Digital computation                     | We discuss why digital physics may be an untestable scientific hypothesis that leads to much more difficult engineering models of the world than the ones we are familiar with                                                                                                                                                                                                                                                                                                                                                                                                | <a href="#">3/18/2023</a> |
| <a href="#">Bob Coecke</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>    | From von Neumann's rejection of Hilbert space to quantum for kids | We'll discuss the shift in conceptual underpinnings that have happened since when John von Neumann wanted an alternative quantum formalism to recent developments, which include unification with causality.                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">3-Mar-23</a>  |
| <a href="#">Jonte Hance</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>   | Using Quantum Technologies to play with Quantum Foundations.      | Jonte discusses two areas of quantum foundations ---interaction-free measurement and (statistical independence-violating) extensions of QM---to show how quantum technologies can be used to probe these.                                                                                                                                                                                                                                                                                                                                                                     | <a href="#">3/4/2023</a>  |

## **Clubhouse: *ItsAboutTime.club***

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                     | CH Title                                                                                  | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Listen                    |
|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">Edward A. Lee</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>       | Computing Needs Time                                                                      | Time in concurrent and distributed software systems is distinctly non-Newtonian and notoriously difficult to reason about. A central challenge is to give system designers an abstraction that they find intuitive while still being able to deliver efficient implementations. One approach is to introduce a logical notion of time, distinct from physical time, that behaves in Newtonian way. This approach exposes some limitations, specifically a fundamental tradeoff between achieving agreement across a distributed system and the time it takes to determine what has been agreed upon. This tradeoff between Consistency (agreement) and Availability (time to access the agreement) gets worse as network Latencies increase, a result called the CAL theorem. A basic goal is to give designers the tools they need to work with this tradeoff. | <a href="#">18-Feb-23</a> |
| <a href="#">Anne Curtis -</a><br><a href="#">Google Scholar</a> <a href="#">ORCID</a>       | Timekeeping in the 21st century - what are atomic clocks and how can we make use of them? | Let's delve into the experimental realities of atomic clocks and explore applications taking them beyond their clock-based origins: navigation, sensing, and testing General Relativity and physics beyond the Standard Model.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">11-Feb-23</a> |
| <a href="#">Johannes Fankhäuser</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | Quantum Uncertainty and Empirical Completeness                                            | Johannes will discuss whether quantum uncertainty is fundamental or if alternative theories could have predictive advantage. The problem connects to locality and reconciles us to quantum features as aspects of limits on Nature's predictability.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <a href="#">4-Feb-23</a>  |
| <a href="#">Barak Shoshany</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>      | Time Travel Paradoxes and Entangled Timelines                                             | Barak will discuss the idea that if time travel is possible, then parallel timelines must also be possible in order to resolve the resulting paradoxes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <a href="#">28-Jan-23</a> |
| <a href="#">Giulia Rubino</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>       | Causality and the arrow of time in the quantum world                                      | Giulia explains how applying key quantum features such as superposition and entanglement to time flow and causal relations bears implications in fundamental physics and benefits for quantum computing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <a href="#">21-Jan-23</a> |

## Clubhouse: *ItsAboutTime.club*

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                               | CH Title                                                                        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Listen                    |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">Lucien Hardy</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>  | <a href="#">Time Symmetry in Quantum Theory: A story with twists and turns.</a> | <p>The story of time symmetry versus time asymmetry in Quantum Theory has undergone various twists and turns. Schroedinger's equation is time symmetric but quantum collapses introduce time asymmetry. Then Aharonov and co-workers found a way to restore time asymmetry for pure state Quantum Theory. However, this was undermined by the Pavia group who showed that causality (the inability to influence the past) reintroduces time asymmetry in the more general case of mixed state Quantum Theory. More recently (in my own work and that of others) it has been seen how to recover time symmetry in Quantum Theory. Is this the end of the story? Is having time symmetry important for the development of a theory of Quantum Gravity?</p> | <a href="#">14-Jan-23</a> |
| <a href="#">Ric Arthur</a>                                                            | <a href="#">The Reality of Time Flow (Local Becoming in Modern Physics)</a>     | <p>Ric Arthur explains how, once temporal becoming is understood to be a local phenomenon, tracked by proper time, and the time symmetry of the laws of physics is properly conceived, all objections to the reality of time flow can be countered.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <a href="#">7-Jan-23</a>  |
|                                                                                       |                                                                                 | <b>Winter Break Christmas Eve + New Year's Eve</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 24-Dec-22<br>31-Dec-22    |
| <a href="#">Michael Frank</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Reversible Computing and Digital Technology Trends</a>              | <p>Mike Frank will talk about the future of digital computing and the need for reversible computing technology, and speculate about why reversible computation may someday be key to saving the universe."</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <a href="#">17-Dec-22</a> |
| <a href="#">Bryan Roberts</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Misfiring Arrows of Time</a>                                        | <p>Our naïve human senses can feel like there is an asymmetry in time when there is not. Here we describe what is needed to have a true arrow of time, in the sense that 'time itself' has an asymmetry.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <a href="#">10-Dec-22</a> |

## [Clubhouse: ItsAboutTime.club](#)

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                           | CH Title                                                                                                                                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Listen                    |
|---------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">Donald Hoffman</a><br><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>        | <a href="#">Fusions of Consciousness</a><br><br>Donald will explore a dynamical theory of conscious agents beyond spacetime, and its projection via decorated permutations onto spacetime. | Donald Hoffman received a PhD in Computational Psychology from MIT, and is a Professor Emeritus of Cognitive Sciences at the University of California, Irvine. He is an author of over 100 scientific papers and three books, including The Case Against Reality: Why Evolution Hid the Truth from Our Eyes. He received a Distinguished Scientific Award of the American Psychological Association for early career research, the Rustum Roy Award of the Chopra Foundation, and the Troland Research Award of the US National Academy of Sciences. His writing has appeared in Scientific American, New Scientist, LA Review of Books, and Edge, and his work has been featured in Wired, Quanta, The Atlantic, Ars Technica, National Public Radio, Discover Magazine, and Through the Wormhole with Morgan Freeman. He has a TED Talk titled "Do we see reality as it is?" and a podcast titled "Reality is an illusion" with Lex Fridman. | <a href="#">3-Dec-22</a>  |
| <a href="#">Mark Van Raamsdonk</a><br><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>    | <a href="#">Entanglement, Spacetime, and Quantum gravity</a>                                                                                                                               | Mark is a Canadian theoretical physicist working at the University of British Columbia in Vancouver after training at Princeton (PhD) and Stanford (postdoc). The main focus of his research is quantum gravity, which he's trying to use to understand black holes and the big bang and to resolve puzzles in cosmology such as "What is dark energy?" Mark is also an amateur jazz musician and teaches "The Physics of Music" in addition to more standard physics courses.                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">26-Nov-22</a> |
| <a href="#">Nicole Yunger Halpern</a><br><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Quantum steampunk: Thermodynamics meets quantum information</a>                                                                                                                | Victorian thermodynamics crystallized the notion of entropy, which quantum information science has extended. Fusing these modern and antiquated sciences, quantum thermodynamics is real-life steampunk.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <a href="#">19-Nov-22</a> |
| <a href="#">Christoph Schiller</a><br><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>    | <a href="#">Physics in 9 lines - Results and Remarkable Consequences</a>                                                                                                                   | 400 years of modern physics are summarized in 9 short lines - 5 principles and 4 sets of choices - that contain general relativity, quantum theory and the standard model of particle physics.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <a href="#">12-Nov-22</a> |
| <a href="#">Sky Nelson-Isaacs</a><br><br><a href="#">Google Scholar</a>                           | The Age of Entanglement<br><a href="#">David Mermin Paper</a>                                                                                                                              | Is the moon there when nobody looks? Understanding entanglement and the 2022 Nobel Prize in Physics. Sky Nelson-Isaacs will explain entanglement through the works of physicist N. David Mermin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <a href="#">5-Nov-22</a>  |

## Clubhouse: *ItsAboutTime.club*

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                     | CH Title                                                                                                                                                                                                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Listen                      |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|
| <a href="#">Clara Aldegunde</a><br>Imperial College London                                  | <a href="#">Spacetime from quantum entanglement: a ticket to unification</a><br><small>This year's Physics Nobel Prize reveals the vital role of entanglement in our Universe, but what if spacetime was essentially threaded by these connections between particles on the quantum level?</small> | This year's Physics Nobel Prize reveals the vital role of entanglement in understanding the Universe, but what if spacetime was essentially threaded by these connections between particles on the quantum level? This hypothesis will be discussed Clara Aldegunde will discuss in this room. Expressing quantum entanglement as a geometry and being able to introduce it to Einstein's spacetime equations is one of the most powerful candidates for a Theory of Everything, reconciling General Relativity and Quantum Mechanics and achieving the ultimate goal of modern physics.                                                                                                               | <a href="#">22-Oct-22</a>   |
| <a href="#">Časlav Brukner</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>      | <a href="#">Quantum Indefiniteness of Causal Relations.</a><br><small>Quantum theory assumes a fixed causal background. Professor Brukner discusses how Indeterminate causal structures enable new quantum information processing system as well as tools for quantum gravity.</small>             | Traditionally, quantum theory assumes the existence of a fixed causal background structure. However, if one applies the laws of quantum mechanics to causal relations, one can imagine situations in which the causal sequence of events is not always fixed, but subject to quantum uncertainty. Such indeterminate causal structures could enable new quantum information processing and provide methodological tools for quantum theories of gravity.                                                                                                                                                                                                                                               | <a href="#">1-Oct-22</a>    |
| <a href="#">Maaneli Derakhshani</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">The Limits of Physics</a><br><small>Some things will forever be beyond physical explanation. In this presentation, we will examine one such thing: basic laws of logical inference, and our ability to reason in accordance with them.</small>                                         | What are the limits, if any, of what can be explained by physics? For that matter, what is physics? It is fashionable among many physicists and philosophers these days to presume that, at bottom, all that exists in the world is 'physical' and in principle can be fully described in physical terms, even if that is beyond our capabilities at present.                                                                                                                                                                                                                                                                                                                                          | <a href="#">24-Sep-22</a>   |
| <a href="#">Mark Burgess</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>        | <a href="#">Causality from a promise theory perspective</a>                                                                                                                                                                                                                                        | Causality has long been a subject of controversy in philosophy, but in physics the meaning is simple: is there a traceable network of influence that maps out a process from start to finish? In the past, physics has viewed causality in a ballistic sense of colliding messengers (billiard balls or canon balls). In the information sciences, this limited view leads us into difficulties, e.g. in quantum mechanics. In most information sciences, the way we consider space and time is based on a cellular model, with interior and exterior processes at different scales. Mark will talk about how this idea of autonomous agents (Promise Theory) can be used to clear up a few mysteries. | <a href="#">17-Sep-22</a>   |
| Summer Break                                                                                |                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3-Sep-22<br>to<br>10-Sep-22 |
| <a href="#">Jonathan Gorard</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>     | <a href="#">A Computational+ Compositional Perspective on the Nature of Time</a>                                                                                                                                                                                                                   | A discussion of how new ideas from applied category theory and the Wolfram Physics Project may help unify distinct notions of time from quantum mechanics, relativity and computational complexity.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <a href="#">27-Aug-22</a>   |

## [Clubhouse: ItsAboutTime.club](#)

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                 | CH Title                                                                   | Description                                                                                                                                                                                                                                                                                                                                                                                                   | Listen                       |
|-----------------------------------------------------------------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <a href="#">David Chalmers</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>  | <a href="#">Reality+. Virtual Worlds and the problems of Philosophy.</a>   | David Chalmers argues that virtual reality is genuine reality. Virtual reality isn't a fiction or an illusion. We can live a meaningful life in a virtual world. We might even be in a virtual world already. He uses this analysis to shed light on coming VR technology and to address traditional philosophical questions about knowledge, reality, consciousness, and value.                              | <a href="#">13-Aug-22</a>    |
| <a href="#">Mark Jeffery</a><br><a href="#">Youtube</a>                                 | <a href="#">Space and time in Wolfram Physics</a>                          | Mark Jeffery talks Wolfram Physics turning our concepts of space and time upside-down. It sees space not as continuous, but as a graph of discrete nodes and edges. And it rescues time from its diminished role as a mere component of space-time.                                                                                                                                                           | <a href="#">6-Aug-22</a>     |
| Break                                                                                   | Wolfram Summer School                                                      |                                                                                                                                                                                                                                                                                                                                                                                                               | 9-July-22<br>to<br>30-Jul-22 |
| <a href="#">Alex Gullen</a><br><a href="#">LinkedIn</a>                                 | Arrival the movie - "A non-zero-sum game".                                 | Alex explores through the lens of the movie 'Arrival' a few key questions and issues that collectively face humanity: are we alone? He'll further discuss a couple key concepts presented in the movie like language, morphology and technology of the Heptapods; and, implications of intelligent extra-terrestrial life like signals in space, the Fermi paradox, the Drake equation, and the Great Filter. | <a href="#">02-07-22</a>     |
| <a href="#">Wisam Reid</a><br><a href="#">Google Scholar</a> <a href="#">Github</a>     | <a href="#">Biological Intelligence: Neurosemantics in Dendritic Space</a> | Wisam will talk about brain matter as the substrate of our intelligence and the most complex tissue in the human body. Dendrites make up more than 90 percent of neural tissue – this "dendritic space" is the cosmos for neuroscientists. Join us as we discuss neurosemantics in dendritic space!                                                                                                           | <a href="#">Jun-18-22</a>    |
| <a href="#">Julian Barbour</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>  | <a href="#">Time, Quantum Mechanics, and the Universe</a>                  | The Founding Fathers of quantum mechanics left the Newtonian conception of time unchanged and sought to explain phenomena observed in the laboratory quite independently of the role that the universe at large could be playing in their generation. Julian will talk about the possible implications of putting the universe at the heart of our intellectual considerations of space, time, and causality. | <a href="#">Jun-11-22</a>    |
| <a href="#">Mohamed Barakat</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Category Theory in Science</a>                                 | Professor Barakat discusses the history of category theory, some of the major developments, and some of the current applications inside and outside of mathematics.                                                                                                                                                                                                                                           | <a href="#">May-28-22</a>    |
| <a href="#">Liane Gabora</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>    | <a href="#">Evolution of Human Conception: Past, Present, and Future</a>   | Liane discusses how our minds became able to (1) recall past events (even in the absence of relevant environmental cues) and weave them into meaningful narratives; and (2) imagine possible futures and how this paved the way for our modern-day conception of time.                                                                                                                                        | <a href="#">30-Apr-22</a>    |

## **Clubhouse: *ItsAboutTime.club***

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                     | CH Title                                                      | Description                                                                                                                                                                                                                                                                                                       | Listen                    |
|---------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <a href="#">Sky Nelson-Isaacs</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>   | <a href="#">Holographic Time in Schrodinger's Equation</a>    | Sky updates current original research on a new way to formulate the time-dependent Schrodinger equation using Fourier transforms. Follow-up on Sky's talk on Spacetime paths as a whole.                                                                                                                          | <a href="#">16-Apr-22</a> |
| <a href="#">Ämin Baumeler</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>       | <a href="#">Time-travel Paradoxes &amp; how to Avoid Them</a> | Ämin discusses the notion of self-consistency, types of paradoxes, and eventually, how to avoid them. This conversation brings us to the question: What is the logical origin of causality, i.e., what feature of nature excludes the possibility of time travel?                                                 | <a href="#">2-Apr-22</a>  |
| <a href="#">Alex Morrise</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>        | <a href="#">String Theory and Beyond</a>                      | Alex asks what is reality? Why is it here? Is there a dynamic explanation for the universe? Is this the only way it could be, are there other worlds possible? How does spacetime emerge from quantum?                                                                                                            | <a href="#">26-Mar-22</a> |
| <a href="#">Mark Burgess</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>        | <a href="#">Smart Spacetime</a>                               | Mark talks about Space & Time and how they have long been the province of theoretical physicists and science fiction writers. Space and time are far from theoretical or fanciful; they are the most practical concern and describe every process in the world—they are always everywhere!                        | <a href="#">19-Mar-22</a> |
| <a href="#">Alex Gullen</a><br><a href="#">LinkedIn</a>                                     | <a href="#">Tenet: The Movie (Time in Science Fiction)</a>    | Alex discusses time, framed by Christopher Nolan's movie Tenet, introducing concepts underpinning this fascinating perspective: time inversion, entropy, temporal pincers, and the Grandfather Paradox. Can the physics of these concepts work? Maybe. Maybe not. At least we are entertained.                    | <a href="#">12-Mar-22</a> |
| <a href="#">Yunus Çengel</a><br><a href="#">Google Scholar</a>                              | <a href="#">Conservation of Information</a>                   | Yunus describes why entropy results in misunderstandings and confusion. The cause: a resemblance of the mathematical formulas in thermodynamics & communications, differing only by a constant factor with the unit 'J/K' thermodynamics & 'bits' in Information Theory.                                          | <a href="#">26-Feb-22</a> |
| <a href="#">Andrei Vazhnov</a><br><a href="#">LinkedIn</a> <a href="#">TEDxRiodelaPlata</a> | <a href="#">Henri Bergson's Philosophy of Time</a>            | Andrei discusses one of the pillars of Bergson's philosophy: time cannot be "spatialized" or quantified. Time may flow unevenly: the past has a fixed and determined character; the future has multiple alternative possibilities, is unpredictable, and is the source of human freedom and creativity.           | <a href="#">12-Feb-22</a> |
| <a href="#">Christoph Schiller</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>  | <a href="#">The Maximum Force in Nature</a>                   | Christoph describes Maximum force implies the first law of black hole mechanics and vice versa. In turn, the first law implies the field equations. It also describes the elasticity of space and determines the maximum shear strength of the elasticity of space. Space elasticity implies the field equations. | <a href="#">29-Jan-22</a> |
| <a href="#">William Sulis</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>       | <a href="#">Locality Is Dead! Long Live Locality!</a>         | Bill discusses the measurement problem and a new intellectual vantage point on Bell's Inequalities from process algebras, yielding insights into the conundrum of non-locality and contextuality.                                                                                                                 | <a href="#">22-Jan-22</a> |

## **Clubhouse: [ItsAboutTime.club](https://itsabouttime.club)**

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                         | CH Title                                                             | Description                                                                                                                                                                                                                                                                                                                                                                                                       | Listen                                            |
|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <a href="#">Paul King</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>               | <a href="#">Neuroscience And The Construction of Reality</a>         | Paul discusses how consciousness and perceived reality likely work in the brain from the perspective of neuroscience: past/memory recall, progression of time, brain's forecasting of the future. View: the neural machinery of the brain is running a real-time virtual reality machine in perceptual information space, and we are living inside its constructed world.                                         | <a href="#">15-Jan-22</a>                         |
| <a href="#">Nick Porcino</a><br><a href="#">Google Scholar</a> <a href="#">Github</a>           | <a href="#">The Mathematics of Time</a>                              | Nick from Pixar talks about the Mathematics of Time. He answers questions: where timing decisions are made and if those choices are procedural or creative; timelines with more than one possible trace; and where, if ever happens, R&D media and entertainment collide with physics.                                                                                                                            | <a href="#">18-Dec-22</a>                         |
| <a href="#">Pauli Erker</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>             | <a href="#">Time, Clocks and Thermodynamics</a>                      | Pauli talks about what is a clock and not a clock, how thermodynamics connects to quantum clocks, and our time-measuring ability's fundamental limitations. What is the relation to Landauer's principle? How can complexity be a resource? A conversation about the experiment with the nano-membrane and the need for autonomous clocks.                                                                        | <a href="#">11-Dec-22</a>                         |
| <a href="#">Wisam Reid</a><br><a href="#">Google Scholar</a> <a href="#">Github</a>             | <a href="#">Neuroscience Connects to Our Experience of SpaceTime</a> | Wisam discusses that a complete unification of theoretical and experimental neuroscience will be required to reveal the neural correlates of intelligence and a general theory of brain plasticity that underlies it. He explores the structure of neural circuits and tests theoretical predictions for plasticity.                                                                                              | 20-Nov-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Mahmoud Jalali Mehrabad</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Roger Penrose's Conformal Cyclic Cosmology</a>           | Mahmoud talks about Penrose's CCC and if it agrees with the second law of thermodynamics. He answers questions such as what happens to the concept of measurement of time at the end of the universe?; how and where can evidence be found for the CCC?; and will CCC, if confirmed, rule out the inflation theory of cosmology?                                                                                  | 6-Nov-21<br><a href="#">Replay Not Available</a>  |
| <a href="#">Julian Barbour</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>          | <a href="#">The Janus Point: A New Theory of Time</a>                | Julian describes time as among the universe's greatest mysteries. Why, when most laws of physics allow for it to flow forward and backward, does it only go forward? To explain this, physicists have long appealed to the second law of thermodynamics, held to predict the increase of disorder in the universe. Julian sees a different possibility that 'order' – the stuff of life – can grow without bound. | 30-Oct-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Mahmoud Jalali Mehrabad</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Roger Penrose's Quantum Consciousness</a>                | Mahmoud explains Penrose's theory on quantum origins of consciousness. "If consciousness arises from a quantum mechanical origin in the brain, what quantum system can be responsible for?" The gravitation of quantum mechanics, quantum entanglement in the brain, the collapse of the wave function in neural network, and why normal digital computers can't model consciousness.                             | 2-Oct-21<br><a href="#">Replay Not Available</a>  |



## Clubhouse: [ItsAboutTime.club](https://itsabouttime.club)

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                                                                                                           | CH Title                                                                       | Description                                                                                                                                                                                                                                                                                                                                                                                      | Listen                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <a href="#">Andrei Vazhnov</a><br><a href="#">LinkedIn</a> <a href="#">TEDxRiodelaPlata</a>                                                                                       | <a href="#">Einstein-Bergson: the nature of time and meaning of relativity</a> | Andrei discusses his interest in Bergson's work and describes him in more detail. Explaining that the central role Bergson played in developing the modern notions of time and his influences on his contemporaries like Poincaré, Lorentz, Michelson, and more. Andrei also comments on how Bergson was responsible for Einstein not receiving the Nobel for relativity.                        | 25-Sep-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Ali Shojaei-Fard</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>                                                                                          | <a href="#">From Infinite Combinatorics to non-perturbative QFT</a>            | Ali discusses infinite combinatorics to non-perturbative quantum field theory. And Feynman's graphon as new analytic graphs, Hopf algebra of Feynman diagrams and topological enrichment, non-perturbative QFT from Dyson-Schwinger equations under different running coupling constants, graphon models of Dyson-Schwinger equations, and the logical foundations of non-perturbative QFT.      | 18-Sep-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Eliahu Cohen</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a><br><a href="#">Marcin Nowakowski</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">Entangled Histories vs. the Two-state Vector Formalism</a>         | Eliahu and Marcin discuss quantum correlations in time and how they generate and are analyzed consistently in the formalisms. They elaborate on a novel behavior of quantum histories of evolving multipartite systems, which do not exhibit global non-local correlations in time but can lead to entangled reduced histories characterizing the evolution of an arbitrarily chosen sub-system. | 11-Sep-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Talesin Beynon</a><br><a href="#">Google Scholar</a> <a href="#">GitHub</a>                                                                                           | <a href="#">Discrete Geometry for Discrete Physics</a>                         | Tali examines his work at the Wolfram Physics project and his interests in discrete and causal geometry, physics, software development (e.g., Mathematica), AI, machine learning, & neuroscience.                                                                                                                                                                                                | 14-Aug-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Adam R. Brown</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>                                                                                             | <a href="#">C, P and T – Are the Law of Physics the Same in Reverse?</a>       | Adam R. Brown responds: What are P-symmetry, C-symmetry, and T-symmetry? What does any of this have to do with the second law of thermodynamics and the arrow of time? Are P, C, CP, T symmetries of nature? Is CPT a symmetry of nature? Why is there more matter than antimatter in the universe?                                                                                              | 24-Jul-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Joscha Bach</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>                                                                                               | <a href="#">Reflections on the Nature of Time &amp; Causality in Cognition</a> | Joscha Bach responds to the relationship between physical time, cognitive time, and subjective time? How is time represented in the brain? How does the subjective observer experience time?                                                                                                                                                                                                     | 17-Jul-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Maaneli Derakhshani</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>                                                                                       | <a href="#">The Reality Problem of Quantum Mechanics</a>                       | Maaneli Derakhshani lectures about "The Reality Problem of Quantum Mechanics." Pre-reading: the concise distinction between the wavefunction of a system vs. universe: Bohmian Mechanics (click on the link in the Title column)                                                                                                                                                                 | 10-Jul-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Flaminia Giacomini</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a>                                                                                        | <a href="#">Entanglement of Quantum Clocks</a>                                 | Flaminia Giacomini's work presents how the gravitational interaction between clocks limits the joint measurability of time at nearby locations. Something similar is found in the Heisenberg uncertainty principle for time, as it is measured by two nearby clocks.                                                                                                                             | 26-Jun-21<br><a href="#">Replay Not Available</a> |

## **Clubhouse: [ItsAboutTime.club](https://itsabouttime.club)**

*A place to discuss our evolving knowledge of the nature of time and causality.  
For physicists, computer scientists, neuroscientists, philosophers and practicing engineers.*

| Speaker                                                                                   | CH Title                                                  | Description                                                                                                                                                                                                                                                               | Listen                                            |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| <a href="#">Taliesin Beynon</a><br><a href="#">Google Scholar</a> <a href="#">GitHub</a>  | <a href="#">Spacetime and the Wolfram Physics Project</a> | Taliesin Beynon discusses his work at the Wolfram Physics project and his interests in discrete and causal geometry, physics, software dev (e.g., Mathematica), AI, machine learning, and neuroscienc                                                                     | 12-Jun-21<br><a href="#">Replay Not Available</a> |
| <a href="#">Sky Nelson-Isaacs</a><br><a href="#">Google Scholar</a> <a href="#">arXiv</a> | <a href="#">The Nature of Time and Causality</a>          | Sky Nelson-Isaacs discusses the mathematical similarities between non-relativistic wavefunction propagation in quantum mechanics. He also explains how image propagation in scalar diffraction theory develops a novel understanding of time and paths through spacetime. | 5-Jun-21<br><a href="#">Replay Not Available</a>  |